

Clear Path Nexus · RailGuard

Decision support for Indian Railways traffic control: build, verification and readiness report

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What this is. A controller decision-support system for the whole Indian Railways network built from real data: a digital twin of every station, junction, section and train in the current all-India timetable on the mapped track; GNSS tracking and live push to every cab; research on where delay is made; freight pathing on the Dedicated Freight Corridors; a re-planner that ranks conflict-free alternatives after a disruption; a driver cab advisory; delay forecasts learned from real running; a tamper-evident audit trail; and a gateway ready for authorised live feeds, tested with a real morning's reports.

What it is not. It is advisory. It has no interface to signals, points, interlocking, Kavach/ATP or brakes, and it does not hold or grant movement authority. Authorised signals, the General & Subsidiary Rules and railway staff remain controlling.

How to read the numbers. The verification in section 2 uses real recorded running of Indian Railways trains (September 2024) and real track data; the safety checks use randomised simulation. None of it is yet a live field trial. Readiness percentages are judgements and are stated with their reasons.

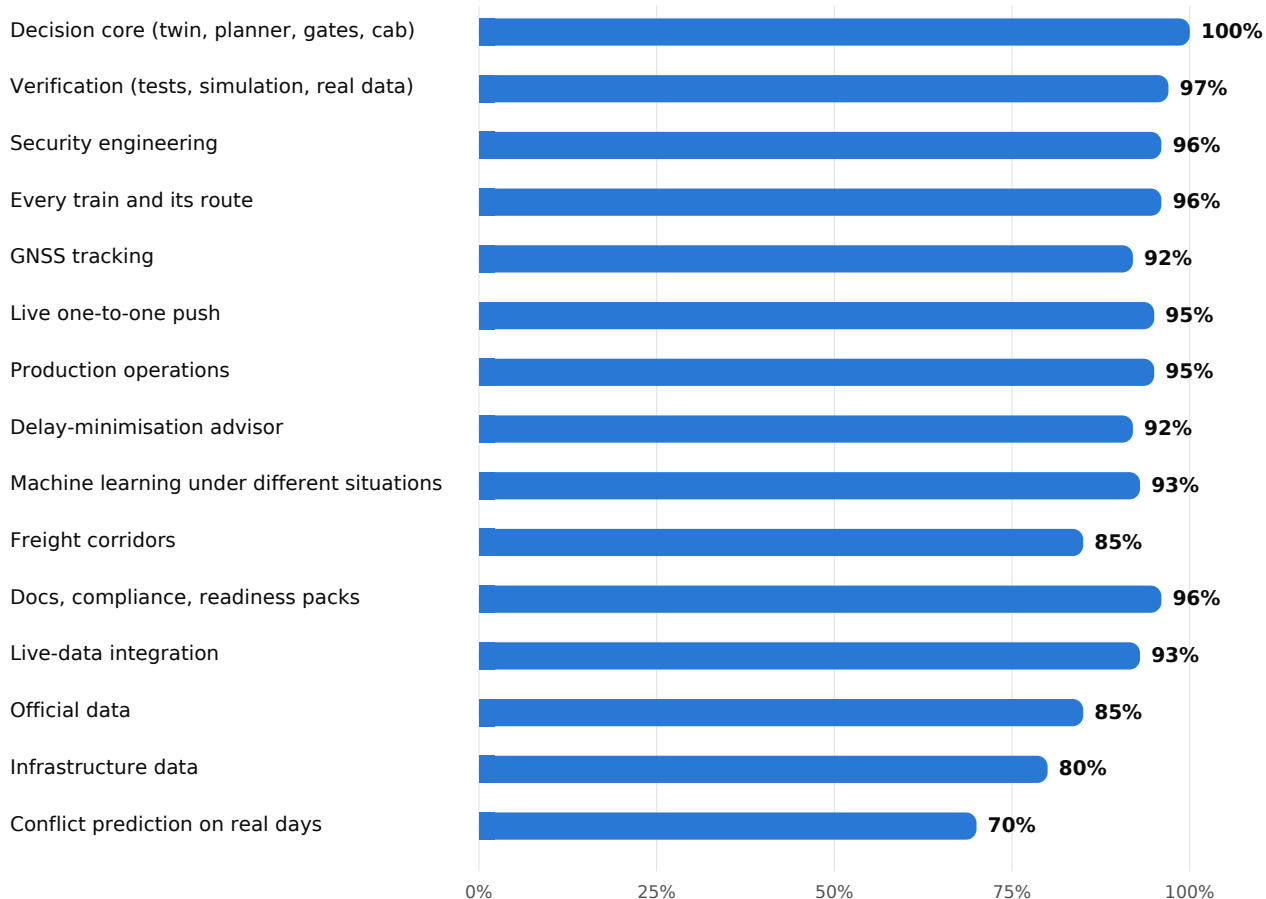
8,550 stations in the twin	3,868 junctions	15,951 sections between stops	10,594 trains in the current all-India timetable
5,068,492 operations checked in simulation	0 invariant violations on the final code	215 automated tests passing	1.38 vs 1.96 min ML error vs section median (locked test)

1. How much is done, and what is left

Software side: about 91% done - everything this project can build and verify, now verified on real recorded running (section 2). The rest of the software work needs inputs only Indian Railways holds: CRIS's interface specification and live feed, block-section and loop registers (what makes conflict warnings precise), the current official timetable, and controller feedback from a trial.

Whole path to railway use: about 58% done, about 42% left. For every railway stage the project has prepared what it can, so that stage becomes review and sign-off. What is left is authorisation, an independent audit, safety acceptance and a field trial. Software alone cannot complete those, so this figure cannot honestly reach 90% before Indian Railways acts.

Software readiness by work package (judgement)

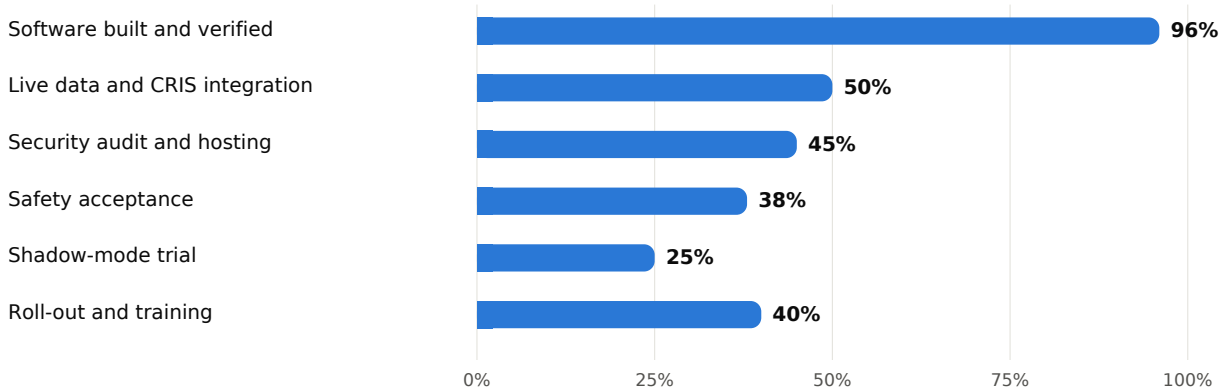


Work package	Done	Why this figure
Decision core (twin, planner, gates, cab)	100%	Built; runs on the current all-India timetable with mapped track; final code verified by simulation
Verification (tests, simulation, real data)	97%	215 tests; 9 invariants incl. GNSS_GATE; 56,395 real train runs; formal V&V; plan needs IR
Security engineering	96%	Named accounts, cab capabilities, hardened container, 41 attack tests, SAST/SCA clean; external audit pending
Every train and its route	96%	10,594 current trains with operator, days, validity, PIN codes and routes on mapped track (90% of sections)
GNSS tracking	92%	Receiver agent, quality gates, map-matching, spoof rejection; real-network check; field units not yet fitted
Live one-to-one push	95%	Console and per-cab streams; 550 streams on the real network, push p50 0.35 s
Production operations	95%	UPS via NUT, signed checkpoints and restore, health, metrics, JSON logs, container restart-tested
Delay-minimisation advisor	92%	Five finding types with levers; 92% of the worst sections recur on held-out days; IR's own data next

Work package	Done	Why this figure
Machine learning under different situations	93%	Rolling-origin rounds; beats baselines in 44/44 situations; cold-start trained; fog/monsoon not in data
Freight corridors	85%	DFC from OSM; 0 conflicts in 200 random days; needs DFCCIL block and loop data
Docs, compliance, readiness packs	96%	Register, 20-hazard log, safety case, deployment guide, operator guide
Live-data integration	93%	Real morning replayed: 96.6% accepted; GNSS agent ready; CRIS mapping remains
Official data	85%	Pipeline ready; data.gov.in refuses this build network; checked against Ministry figure
Infrastructure data	80%	OSM line count on 90% of sections; block sections and loops need IR registers
Conflict prediction on real days	70%	Warnings 1.23x as likely to precede real time loss; needs IR block data

Path to railway use (weights are shares of total effort)

Done per stage (judgement)



Stage	Weight	Done	Done by this project	Left: only the railway can do
Software built and verified	35%	96%	This repository, verified on real running data and the real network	-
Live data and CRIS integration	15%	50%	Signed gateway proven on a real morning; GNSS cab agent; live push to cabs; contract; test vector	Authorise access; share spec and block-section data; issue keys; fit cab units; test in CRIS environment
Security audit and hosting	10%	45%	ASVS self-assessment, 41 attack tests, SAST/SCA, named accounts, hardened container restart-tested	CERT-In/STQC audit; host on IR infrastructure; connect IR identity
Safety acceptance	15%	38%	Hazard log (20 hazards) and safety case with real-data evidence	Hazard workshop; scoring; independent assessment; RDSO/IR acceptance
Shadow-mode trial	20%	25%	Retrospective shadow run on real days; trial tooling, protocol, metrics	Run 3-12 months on a division with live data; review disagreements
Roll-out and training	5%	40%	Console with sign-in and live push; operator and deployment guides	Deliver training; issue procedures

Readiness packs: seva2026/railway_readiness/ - hazard log, safety case, live-data interface contract, ASVS checklist, shadow-trial plan, operator guide.

2. Verified on real running

Everything in this section is computed from what really happened: the actual arrival times of 56,395 train runs (1,262,418 station arrivals, 3,549 trains) in September 2024, published for research by Chowdhury, Koley, Chakraborty and Ghosh (IIT Kharagpur, IEEE T-ITS 2026). Real track data comes from OpenStreetMap. Nothing here is generated by this project. The running data is fetched by the user, checked against a pinned SHA-256 and never redistributed; only these aggregate results are published.

56,395

real train runs checked

1.26M

actual arrival times

8,188

sections with mapped
track data

13.7 min

forecast error on unseen
days

Is the data credible?

Measure	Observed (Sep 2024)	Official
Mail/Express at destination within 15 min	82.0%	77.12% (FY 2024-25, Ministry of Railways)

82.0% on time in September 2024 against the Ministry's 77.12% for the whole financial year, which includes the winter fog months: a credible sample of real running.

What the real data changed in the twin

Item	Before (open 2016 data)	Now (real data)
Timetable	2016 community snapshot	September 2024: 3,549 trains, 78,892 timed stops
Running days	unknown for every train	from observed running for all 3,549 trains
Station positions	community coordinates	8,224 located on the map by their IR code
Track count	inferred from timetable or assumed single	mapped on 8,188 of 9,065 sections: 3,862 single, 4,325 double or more
Electrification	unknown	7,830 sections at least 90% electrified
Trains with placeholder times	not detected	343 excluded and counted

Of the trains running under the same number, origin and destination in 2016 and 2024, 64.2% changed journey time by more than 15 minutes, and 6,091 sections of 2024 did not exist in the 2016 data: running on the current timetable matters.

Forecasting where late trains will be (forward in time: learned on 1-20 Sep, scored on 21-30 Sep)

Method	Average error	Within 15 min	<=1 h ahead	1-3 h	3-6 h	>6 h
Twin's previous rule	17.2 min	73.2%	9.0	15.0	19.8	30.0
Delay stays the same	18.4 min	70.2%	8.6	15.1	20.9	35.2
Train's own history	15.8 min	75.8%	7.0	12.2	17.8	31.9
Learned from real running	13.7 min	80.5%	6.3	11.0	15.4	26.7

1,415,666 forecasts for 18,754 real runs on days the model never saw. Improvement over the previous rule: 3.52 min (95% interval 3.39-3.66, bootstrap over runs). The P10-P90 band covered 81.6% of real outcomes (target 80%). The deployed model is exactly the one scored here; it is rebuilt locally and not distributed.

Do conflict warnings come true? (retrospective shadow run on real days)

At six times of day on four unseen real days, the national twin was given every train's real delay and flagged conflicts in the next two hours. For each warning, the train that has to give way should then lose time on that section. It is compared with unflagged trains in the same state (already late or not), because late trains tend to recover on their own.

Twin	Warnings	Give-way train lost >=5 min	Comparable trains	Lift
Previous projection, no track data	1,907	26.43%	25.46%	1.04x

Twin	Warnings	Give-way train lost >=5 min	Comparable trains	Lift
Previous projection + real track data	1,461	27.24%	25.45%	1.07x
Learned projection + real track data	1,603	29.88%	23.9%	1.25x

Real track data removed 493 of 839 single-line warnings that the old 'assume single line' rule raised. Warnings are informative but far from certain: section data between stopping stations cannot see block sections, intermediate loops or signals. That data is held by Indian Railways (engineering registers, COA) and is the next real gain.

A real morning through the signed live-feed gateway

Projection	Reports	Accepted	Batch p50 / p95	Projection within 15 min of reality
Previous rule	7,979	96.6%	126 / 162 ms	76.4%
Learned forecast	7,979	96.6%	162 / 218 ms	80.9%

06:00-10:00 IST on 2024-09-24: every real arrival sent as a signed NTES-style STATION event at its real time. Rejected reports are out-of-order or unknown runs, refused on purpose.

Defects found by real data (all fixed)

Finding	What changed
Live-feed gateway threw away real reports	A late train that had not reported yet was projected on time, so its real report at a station the projection had already passed was rejected (19% of a real morning's reports). Reports are now matched from the train's last accepted observation, never from the projection; a report behind it is refused as out of order.
Threats re-evaluated once per event	Every report triggered a national threat evaluation: 0.5 s per one-minute batch in the first hour. Now once per batch: about 0.1 s over a whole morning, and every change still supersedes earlier rankings.
Placeholder timetables in the source	343 trains (mostly specials) list every stop at the same minute; their 'actual' times mean nothing. They are excluded from the twin and from every score, and counted.
Track count over-read on curves and corridors	The first OSM line count read curving single lines and the parallel Dedicated Freight Corridor as double line. It now counts lines crossing a perpendicular cross-section, ignores freight corridors, metros and sidings, and was checked against known single (Konkan) and double (Delhi-Bhopal) routes.
Old timetable and stale labels	The 2016 timetable lacked 6,091 sections of 2024; the twin now runs on the 2024 one, and its data label no longer claims the network is inferred from the open timetable.
Conflict warnings barely predictive with the old projection	Carrying a delay forward unchanged made warnings only slightly better than chance; projecting late trains with the forecast learned from real running made them clearly more predictive (see table).
Forecast plan overlapped itself (found by the simulation)	A forecast recovering time faster than possible made a plan leave a stop before arriving (7 of the first 60 episodes). Forecast plans now never do that and never run a section under 85% of its timetabled time.
A forecast could replace a controller's decision	A forecast now only replaces a projection; once a hold or yield shapes the plan, new delays are added on top.
Simulation workers stalled	Model inference threads in every worker spun against each other; now one thread per worker (600 episodes in 35 s).

Limits. One month (September: no fog season); only arrivals are observed (departures in the file are arrival plus scheduled halt); early running is recorded as zero; freight and suburban trains are not included; OpenStreetMap is volunteer-mapped. A shadow trial on live data with controllers is still needed.

3. Production build: every train, live tracking, operations

This build takes the twin from the 2024 timetable to every train running now, tracks trains by GNSS on the real track, pushes changes live to each cab, adds research on where delay is made, plans freight on the Dedicated Freight Corridors, and adds what a control centre needs to run it: named accounts, UPS power handling, state checkpoints, health and metrics, and a hardened container.

Every train and the track it follows

Measure	Value
Trains in the current all-India timetable (valid 30 Aug-30 Sep 2026)	10,594
Train numbers known across all sources (current, 2024, 2016)	13,451
Operated by IRCTC (private), Bharat Gaurav, luxury tourist, parcel, rapid rail	Parcel / cargo: 6, Rapid rail: 4, IRCTC (private operation): 4, Bharat Gaurav (service provider operated): 3, Luxury tourist train: 1
Current trains with observed real running (for forecasts)	2,977 (28%)
Stations / with a PIN code (approximate, OpenStreetMap postcodes)	8,550 / 5,073
Sections with the real mapped track geometry	14,228

Each train's route is returned as a line along the mapped track (GeoJSON), with the kilometres on mapped track, on single line and electrified, and every section without a mapped path labelled as drawn straight. Passenger booking records (PNR) are personal data held by IRCTC/CRIS and are not collected.

GNSS tracking on the real network

Every train running at 10:00 on the current timetable reported a fix each minute for ten minutes, from simulated receivers (HDOP 0.6-2.5, 2% multipath of 30-80 m) on the real mapped track; 5% of fixes were replaced by attacks. The receivers are simulated because no live GNSS feed is available; the network, timetable and track are real.

Measure	Result
Genuine fixes accepted	21,777 of 21,851 (99.661%)
Matched to mapped track (rest: sections without mapping)	90.9%
Genuine fixes refused quietly (multipath, no alarm) / raising an alarm	66 / 4 (0.02%)
Jumps of 15 km in 60 s refused	100.0%
Teleports refused	100.0%
Fixes 500 m off the track refused	90.21% (the rest lie beside sections without mapped track, matched with 3 km slack)
Gateway throughput	2,792 fixes per second

Live, one-to-one push

Measure	Result
Concurrent streams held (control screens + cab units, own tokens)	50 + 500; dropped 0, refused 0
Controller decision to the affected cab (push latency)	p50 352 ms, p95 1311 ms (n=9)
First event after connecting	p50 1420 ms
Signed feed batch of 250 fixes, while streaming	p50 159 ms
Network	2,537 trains running; 4 CPUs; separate process, one uvicorn worker

Delay-minimisation advisor

From real running (30 days, 1,198,313 station-to-station traversals): the network loses 183,355 train-minutes a day and recovers 157,240. Findings are made on 1-15 September and must recur on 16-30 September: 92.0% of the worst 50 sections and 94.0% of the worst 50 stations do. Incidents over 3 hours and schedules that do not describe a train's running are counted apart, not mixed into the patterns.

Finding	Count	Lever
Chronic section losses (top 200)	178	Single line: crossing loops / doubling; double line: restrictions, block spacing, precedence
Trains starting late on most days	140	Rake links, pit-line slots, crew booking
Timings no train achieves (loses 5+ min on 3 days in 4)	3829	Re-time, moving allowance from where the train always recovers
Chronically late trains (on time on fewer than half the days)	226	The train's own list: worst sections and start

Forecasts under different situations

Retrained in 5 rolling-origin rounds (each learns only from days before its origin and is scored on the next four). Average error: persistence 19.1 min, the twin's rule 17.88, learned 14.19. The learned forecast beats both baselines in 44/44 situations (time of day, weekday, current delay, train class, line type, horizon, zone, network disruption, history). With all history removed (a new train) it scores 16.04 min against 19.1 for persistence, and reported delays off by +/-3 min cost it 0.05 min. The deployed model is trained with that cold-start case. Fog and monsoon are not in a September sample.

Dedicated Freight Corridors

Corridor	Random days planned	Trains	Conflicts in independent check	At published daily volume: mean wait / on free path
Eastern	100	18,002	0	295.3 min / 24.4%
Western	100	15,473	0	14.1 min / 63.2%

Random demand is uniform along the corridor, which loads the Eastern corridor's single-line stretch (Ludhiana-Khurja) harder than real traffic; DFCCIL block and loop data and FOIS demand replace it.

Operations

- Power: a UPS watched through Network UPS Tools. On battery every console shows POWER_ON_BATTERY and state is checkpointed every 10 s; on low battery approvals pause and readiness fails so users move to the standby.
- State: signed, atomic checkpoints, restored on start only if authentic, of the same data and recent; the container restored its state after a restart.
- Health and monitoring: /health/live, /health/ready, Prometheus /metrics; JSON access logs carry the route template only, never secrets (checked on a production container).
- People: named accounts with lockout and hashed sessions; every decision records the signed-in person.
- Container: non-root, read-only root filesystem, capabilities dropped, digest-pinned base image; deployment guide in seva2026/railway_readiness/DEPLOYMENT.md.

4. What was built

Component	What it does
National twin (railguard/national.py)	Every station, junction and section of the 2024 timetable with mapped track data; train runs over a two-day window on their observed running days; occupancy index for fast conflict search; disruption, ranking, approval, threats, cab, replay.
Real data (realdata.py, osm_infra.py, realval.py, railguard/eta.py)	Observed running and the 2024 timetable; OpenStreetMap track data; verification against what happened; delay forecasts learned from real running for feed-reported trains.
Tabletop twin + TwinTrack (engine.py, hardware/)	A 10-section demonstration line with two trains, ESP32 sensor/cab firmware and six scripted scenarios.
Planner and scoring (planner.py, scoring.py)	Generates CONTINUE / HOLD / PATH-THROUGH / PRIORITY (with cascading re-paths) / REROUTE options, resolves conflicts by making lower-priority trains yield, ranks by 7 explainable factors.
Evidence gate and approval	Fresh/aging/stale evidence states; a named controller approves only the latest recommendation for the current state; anything older is superseded.
Threat registry	Converging paths, opposing occupancy, blocked/closed sections, stale data, route deviation, impossible movement; OPEN → ACKNOWLEDGED → CLEARED lifecycle.
Nexus Control / Nexus Cab (static/)	Controller console (tabletop and national map) and driver cab view; strict CSP, no inline code, tokens never in URLs sent to the server.
Audit (audit.py)	Hash-chained events with optional HMAC, persisted as JSON lines, verified by CLI; decision snapshots that replay to the same ranking.
Run-time ML (runtime_model.py, training.py)	Gradient-boosted trees with quantile intervals and conformal calibration; measured training rounds with a locked test set.
Official data (official.py)	Registry, licences, SHA-256 provenance and a parser for the data.gov.in timetable; official rail distances replace straight-line estimates when present.
Live-feed gateway (livefeed.py)	Signed batches from authorised sources (RTIS/NTES/COA-style); replay protection; GNSS map-matching; station events become disruptions for the controller.
Security layer (security.py)	Roles, fail-closed production mode, rate limits, body caps, host allow-list, strict request schemas, security headers.
Working schedules (schedules.py)	Every train's stop matrix (day, times, dwell, distance, section speed), per-train metrics and a completeness audit of all 5,208 trains; running days honoured when official data supplies them.
Shadow trial (shadow.py)	Logs controllers' actual decisions beside the recommendations shown at the time and measures agreement, without changing anything: the evidence a field trial needs.
Simulation harness (simulate.py)	Randomised operation sequences on both twins with 8 safety invariants checked after every operation.
Every train (current.py, registry.py)	The current all-India timetable (10,594 trains) with operators, running days and validity; a registry across all sources; station PIN codes; each train's route along the mapped track.
GNSS tracking (railguard/gps.py, gnss_agent.py)	NMEA (GPS/NavIC) parsing, quality gates, map-matching onto mapped track with yard and doubled-back track handling, spoof and jump rejection, and the cab unit's signing agent.
Live push (railguard/live.py)	Server-sent event streams for consoles and each cab, one shared computation per change, run-scoped cab capability tokens, heartbeats, bounded capacity.
Operations (railguard/ops.py, Dockerfile)	UPS through NUT, signed checkpoints and restore, health and readiness, Prometheus metrics, JSON logs; hardened container and compose file.
Accounts (accounts.py)	Named users with scrypt passwords, lockout, hashed sessions and roles; decisions carry the person.
Delay advisor (delay_advisor.py) and scenario training (scenario_ml.py)	Where delay is made and the lever for each finding, checked for persistence; forecasts retrained and scored by situation in rolling-origin rounds.
Freight (freight.py)	Dedicated Freight Corridors from OpenStreetMap and conflict-free freight pathing with an independent check.

Component	What it does
Assistant (assistant.py, free_assistant.py)	Offline or local open-source model (Ollama) by default; the paid Claude option is opt-in only.

Technology used

Area	Technology
Language / runtime	Python 3.11 (CI) and 3.13; HTML, CSS and plain JavaScript (no framework); Arduino C++ for the ESP32
Service	FastAPI 0.142, Starlette 1.7, Pydantic 2 (strict models), Uvicorn
Data	SQLite, pandas, NumPy, SciPy; observed running Sep 2024 (research use); OpenStreetMap (ODbL) via pyosmium; DataMeet (CC0); data.gov.in pipeline (GODL)
Machine learning	scikit-learn HistGradientBoosting (absolute-error and quantile losses), GroupKFold, split-conformal calibration, joblib
Security	HMAC-SHA256, SHA-256 hash chains, constant-time comparison, token buckets, CSP; Bandit (SAST), pip-audit (SCA)
Quality	pytest, ruff 0.8.6 (pinned, as in CI), radon, vulture, GitHub Actions CI, Playwright + Chromium for UI checks
Assistant	Ollama (local, free) or the Anthropic SDK (opt-in, paid)
Report	ReportLab (this PDF)

5. Data: the whole network

The twin is built from real data: by default the current all-India timetable (10,594 trains with running days and validity dates, section 3), with station positions, line counts and track geometry from OpenStreetMap; the September 2024 timetable with observed running days (3,549 trains; 343 placeholder timetables excluded) is what the system is verified against (section 2). Every attribute is labelled with its evidence, so nobody mistakes an inference for a survey. Production refuses the 2016 open timetable. The figures below are for the timetable the twin runs on.

Quantity	Value	How it was obtained
stations total	8,550	every station in the open and 2024 data
stations located from osm	7,889	positioned on the map by their Indian Railways code (OpenStreetMap ref)
stations	8,550	stations where at least one train halts
stations without halt placed on sections	0	non-halt stations projected onto the section they lie on (≤ 2 km)
stations without halt not placed	0	no coordinates, or farther than 2 km from every section
junctions	3,868	stations joined to 3 or more neighbours
sections	15,951	pairs of consecutive stops
sections with osm infrastructure	14,228	mapped track path accepted (agrees with the timetable's rail distance)
sections osm double or more	4,078	two or more parallel running lines mapped along the section
sections osm single	7,485	a single-line stretch over 2 km, or over 10% of the section
sections inferred multi track	4,829	opposing trains timetabled on it at the same time (they must cross inside it)
sections assumed single	1,291	no evidence at all: treated as single line (the safe assumption)
runs in window	9,861	train numbers x start days in 2 days
occupations	193,554	section occupations in the window

Official data. india_rail/official.py registers the Government sources (data.gov.in timetable, Trains at a Glance, station codes) with publisher and licence. It parses the data.gov.in format, including its 00:00:00 placeholders, and records each file's SHA-256 in official_provenance. It reconciles the file against the community data and gives the twin official rail distances per section. The Government

portals reset connections from this cloud build network, so the file must be fetched from India with `python -m india_rail official --file <csv>`. Tests cover the whole path with a format fixture.

Known limits. The timetable is from September 2024 (the current official one must be ingested from India). OpenStreetMap is volunteer-mapped. Loops, platforms, block sections and signals are not in any open dataset; they are what conflict warnings need next (section 2).

All trains in the open data: working schedules and completeness

Every published train was checked: `python -m india_rail schedules --audit`. Each train's working schedule (every stop with day, arrival, departure, dwell, distance from origin, section run time and speed) is available from `python -m india_rail schedules --train <number>` and `GET /trains/<number>/working-schedule`.

Measure	Value
Trains in the published list	5,208
With a usable timetable	5,187 (99.6%)
Timed stops / halts with dwell	394,125 / 93,620
Timetable duration within 30 min of the published duration	5,129 trains (median difference 0.0 min)
Published duration / distance known	99.71% / 99.19%
Accommodation classes / return train known	45.87% / 88.5%
Running days known	0.0% (not in the open data)
Stations with suspect published coordinates	105 (e.g. KRLS 91, GOH 91, KTHL 87, MIR 87 impossible sections)

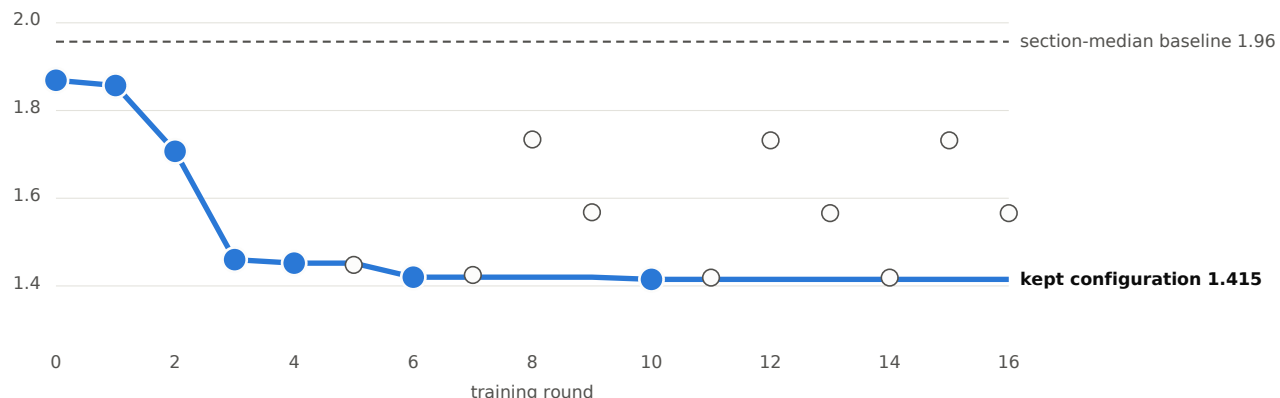
Flag	Trains
RUNNING_DAYS_UNKNOWN	5,208
NO_CLASS_INFORMATION	2,819
STOPS_WITHOUT_COORDINATES	1,071
DISTANCE_INCONSISTENT_WITH_TIME	794
DURATION_DIFFERS_FROM_PUBLISHED	50
NO_PUBLISHED_DISTANCE	42
RETURN_TRAIN_NOT_LISTED	38
NO_USABLE_TIMETABLE	21
NO_PUBLISHED_DURATION	8

6. Machine learning: section run-time model

The model predicts the scheduled run time between two stops: the planning quantity a re-planner needs. It learns the correction to a section prior. Section priors are recomputed inside every cross-validation fold from training trains only, and folds are grouped by train pair, so no evaluated train is ever seen in training.

Training protocol. Before any training round, 15% of train groups were locked away as a final test set (58,110 sections, 767 trains). Each round changed one thing and was kept only if 3-fold grouped CV error improved by at least 0.005 min. Training stopped when a full pass found no further gain.

Cross-validated mean absolute error by round (minutes; lower is better)



Round	Change	Kept	CV MAE (min)	Within 2 min
0	baseline (current production configuration)	yes	1.869	76.0%
1	round predictions to whole minutes (timetables are minute-resolution)	yes	1.857	81.7%
2	type- and direction-specific section priors, run-time spread on the section	yes	1.707	83.6%
3	local pace from the two sections either side (existing trains)	yes	1.460	87.4%
4	start from the type-specific section prior	yes	1.452	87.3%
5	more trees, slower learning	no	1.448	87.4%
6	larger trees	yes	1.420	87.7%
7	smaller leaves, more regularisation	no	1.425	87.7%
8	blend toward well-supported section medians (k=20)	no	1.734	84.3%
9	blend toward well-supported section medians (k=60)	no	1.568	86.4%
10	more trees, slower learning	yes	1.415	87.8%
11	smaller leaves, more regularisation	no	1.419	87.7%
12	blend toward well-supported section medians (k=20)	no	1.732	84.3%
13	blend toward well-supported section medians (k=60)	no	1.566	86.5%
14	smaller leaves, more regularisation	no	1.419	87.7%
15	blend toward well-supported section medians (k=20)	no	1.732	84.3%
16	blend toward well-supported section medians (k=60)	no	1.566	86.5%

Locked test set, scored once

Predictor	MAE (min)	Within 2 min	MAPE ≥ 10 -min sections
Final model	1.378	88.1%	16.0%
Section-median baseline	1.957	80.1%	24.4%
Speed-by-train-type baseline	2.611	63.3%	30.7%

Production model (5-fold grouped CV on all data)

Mode	MAE (min)	Within 2 min
Existing train (uses its other sections)	1.387	88.0%
New path (no train context)	1.680	83.7%
Section-median baseline	1.953	80.4%

P10-P90 interval coverage after conformal calibration: 79.99% (target 80%), mean width 3.855 min.

Why not 'perfect'. The target is a published timetable in whole minutes with operator-chosen allowances that no public feature explains, so zero error is not achievable. The real gain comes from actual running: section 2 trains a delay forecast on 1.26 million real arrivals and scores it on unseen days; with IR's live feed it can be retrained across seasons.

7. Simulation: 11.8 million checked operations

An episode starts a fresh twin with random conditions, then runs a random sequence of operations. The operations are clock ticks, recommendations, approvals (including stale and wrong ones), section and sensor events, malformed or impossible feed data, holds and acknowledgements. Eight safety invariants are checked after every operation (a ninth, GNSS_GATE, was added with GNSS tracking), and every episode is seeded so any failure can be replayed exactly. Full end-to-end episodes cost about 0.1 s each, so the 5-million target is counted honestly as checked operations, not as episodes.

Result. The main run checked over 5 million operations. The national twin had no violations. The tabletop twin had 8 violations in 70,000 episodes: 7 were one real defect (re-holding a train that was already held released it forward) and 1 was a test-case error in the harness. Both were fixed, all 8 episodes replay clean, and a further run on the fixed, final code is reported below.

Twin	Episodes	Operations checked	Violations	CPU ms / episode
demo	70,000	4,018,172	8	135.303
national	90,000	1,050,320	0	104.996
Total	160,000	5,068,492	8	

Simulated 5-second steps (tabletop twin): 55,375,197. Wall time 105 min on 3 workers. Code checksum 52f194c00c9a6c06...

Regression run (after the main run's code): 12,000 episodes, 411,434 operations, 0 violations (code checksum 34e99e011fc49a26...).

Run on the code before real data: 24,000 episodes, 1,201,001 operations, 0 violations (code checksum 752bfcc1c5691bb8...).

Final code on real data with the learned forecast: 36,000 episodes, 960,020 operations, 0 violations (code checksum 55bda8517626f039...).

Final code, national twin on the real network: 200,000 episodes, 2,227,905 operations, 0 violations (code checksum 55bda8517626f039...).

Production build, first run: the new GNSS_GATE invariant found one defect (fixed, section 9): 36,000 episodes, 957,617 operations, 1 violations (code checksum 307a4057aace267f...).

Production build: final code, both twins, GNSS fixes through the signed gateway: 36,000 episodes, 962,191 operations, 0 violations (code checksum b349b021ce92b5a5...).

All runs together: 11,788,660 checked operations. Each of the 9 violations was traced to a defect (or, once, a harness error), fixed and replayed clean; the final code has 0.

Invariants

Invariant	Meaning
SAFE_SEPARATION	Approved plans never conflict; simulated trains never share a single-line section or close up on one track; later conflicts entering the planning horizon are always on the threat list
APPROVAL_GATE	Only the latest recommendation for the current state can be approved, with fresh evidence, no open critical threat, and no closed or obstructed section on the plan
EVIDENCE_GATE	Stale or missing mandatory evidence never yields an approvable recommendation
INPUT_REJECTION	Malformed, replayed, future or physically impossible observations never move the twin
CAB_ADVISORY	No speed band without fresh evidence and an approved plan; never a proceed status with a blocked section ahead; band never above line speed
RANKING	Ranked plans are conflict-free, feasible, monotonic in time, never early, scored 0-1, winner first; yields never re-route a train
REPLAY	Snapshots verify and replay to the same ranking; the audit chain verifies
GNSS_GATE	Through the signed gateway, a GNSS fix that jumps along the route faster than any train, lies far off the route or has too few satellites is never accepted and never changes the train's position evidence
NO_CRASH / LIVENESS	No unexpected exception; every episode completes within 60 s

What the simulation found (all fixed, each with a regression test or replayable episode)

#	Finding	Fix
1	An approval went through after the state changed at the same clock time (e.g. a controller hold), and cancelled the hold	Every state change bumps a version; approval must match the ranked version
2	A train re-planned mid-section was moved back to the start of the section	Plans carry the starting offset
3	A held train could stop at the start of the next section instead of the station	Holds stop at the node
4	A closed section ahead raised no threat on the tabletop twin (cab showed NORMAL)	SECTION_CLOSED critical threat
5	Rounded plan times put a re-planned train back at the station for a fraction of a minute	Position held until the plan's first time
6	National planner checked closures and conflicts only from the disruption station onward	Planning starts at the first section the train can still change
7	PRIORITY-path option skipped the closed/obstructed/axle-load check	One feasibility filter for every option
8	Conflicts on the section a train already occupies were never examined	Conflict search starts at the occupied section
9	Two re-pathed trains could ping-pong forever (planning hang)	Bounded cascade; LIVENESS invariant
10	The 240-min horizon was applied per train, so a pair was seen from one side only	Pair-symmetric horizon
11	The PRIORITY option returned a plan that its own re-check had not validated	Return the validated plan
12	Replay did not reproduce rankings (threats/acks not restored; order-dependent index)	Restore derived state; sorted index; 1,585 replays 0 mismatches
13	A disruption reported by a live station event could not be applied where the train really was	Feed events pin the stop index
14	Main run, 7 episodes: re-holding an already-held train released it towards its timetabled position	A re-hold never moves the stop point later; regression test fails on the old code
15	Main run, 1 episode: a harness test case (not the system) mislabelled a plausible move as a jump	Harness fixed

8. Security

Check	Result
Adversarial API tests	41 cases pass: auth bypass, token tricks, path traversal, NaN/Infinity, type confusion, JSON bombs, injection strings, method tampering, CORS, error leakage, role separation
Security tests	Fail-closed production mode, role separation, host allow-list, body limits, security headers, rate limits, bounded memory, audit tamper detection
Live-feed tests	Bad signature, unknown key, replayed nonce, stale timestamp, sequence regression, oversize batch, off-route fixes
Static analysis (Bandit)	0 findings (18 reviewed suppressions, each justified in the code)
Dependency audit (pip-audit)	0 known vulnerabilities in 49 pinned packages
Browser	Strict Content-Security-Policy verified in Chromium: no console errors on any page

Security findings fixed during the audits

#	Finding	Fix
1	Starlette 0.41.3 had 7 published advisories; pytest 8.3.4 had 1	FastAPI 0.142.2 / Starlette 1.7.0 / pytest 9.0.3; pip-audit 0
2	NaN in a request crashed the 422 response (500) and errors echoed client input	Validation handler that never echoes input
3	Lax coercion accepted "5" and true as numbers and ignored unknown fields	Strict request models: exact types, no extra fields, finite numbers
4	Audit chain could not be verified after a restart (second GENESIS in one file)	Chain resumes from the file
5	Ollama URL from the environment was opened without a scheme check (file://)	http(s) only
6	SQL assembled with an f-string (safe but fragile)	Constant statement, all values bound
7	Chunked uploads bypassed the body-size cap	411 for bodies without a length
8	Interactive API docs exposed in production	Disabled when RAILGUARD_MODE=production

9. Seven audit passes

Each pass looked at the whole system through one lens, fixed what it found, and re-ran the full test suite and lint before the next pass. Details are in [seva2026/AUDIT_LOG.md](#).

Pass	Lens	Main findings and fixes	Evidence
1	Safety logic	14 system defects found by the randomised simulation and fixed (section 7): stale approvals, position snapping, holds and re-holds, closures, planner scope, cascade hang, horizon symmetry, replay determinism	simulate.py; 0 violations on the final code
2	Security	8 findings fixed (section 8): dependency CVEs, NaN crash and input echo, type coercion, audit restart, URL schemes, SQL construction, chunked bodies, docs exposure	test_attacks.py, Bandit 0, pip-audit 0
3	Performance	Plan and stress caches keyed on exact inputs, cheap snapshot copies, distance cache: 39% less CPU per episode with byte-identical outputs (6 scenario and 30 episode checksums equal)	baseline checksums, profiles
4	Data and ML	Locked-test training protocol; model error 1.96 → 1.38 min (-30%); production model retrained; official-data pipeline; 914 non-halt stations placed; placeholder-time bug caught by tests	training_log.json, test_official.py
5	Code quality	Dead code removed; national builder split (complexity F 62 → C 16) with an identical data checksum; lint/format clean on Python 3.11 and 3.13; CI parity checked	radon, vulture, ruff, CI
6	Real-life data	10 findings fixed (section 2): feed reports rejected by a projection, per-event threat refresh, placeholder timetables, OSM line over-count, old timetable, weak warnings, forecast plan overlap (caught by the simulation), forecasts replacing decisions, worker thread contention, stale labels	56,395 real runs; realval.py; REAL_DATA_VALIDATION.md
7	Production build	13 findings fixed (section 3; one found by the simulation): GNSS matched to the wrong pass of curving or reversing track (implausible refusals 99 → 4), stopped trains drawn on the station pin off the track, straight-line slack swallowing off-track fixes, one multipath fix raising an alarm, silent live streams, per-stream polling (push p50 0.96 → 0.35 s), a lint auto-fix breaking a pandas group-by, rate-limit state leaking between tests, trains off their timetable distorting the advisor (persistence 78% → 92%), forecasts weak for trains without history (72% of current trains), image build behind a TLS-inspecting proxy, scanner findings in the load-test harness, a fix between two possible places on shared track accepted (GNSS_GATE)	gnss_verification.json, loadtest.json, scenario_ml.json, GNSS_GATE

10. Ready for live data

When the Ministry of Railways / CRIS authorises a feed, it plugs in without code changes to the twin.

- Envelope: source, key id, sent-at, nonce, sequence and events, HMAC-SHA256 signed with a per-source key (RAILGUARD_FEED_KEYS). Refused before reading: bad signature, unknown key, clock skew beyond ± 120 s, replayed nonce, non-increasing sequence, more than 250 events.
- POSITION events (RTIS or cab GNSS units) pass receiver quality gates and are map-matched onto the mapped track of the train's planned sections (50 m or 3x the stated accuracy, 300 m in station yards; 3 km beside unmapped sections). A fix that jumps or runs backwards is refused as GNSS_IMPLAUSIBLE; one far off the route raises a route deviation at once, a near one on the second in a row.
- Cab GNSS units run the device agent (python -m india_rail gnss) with their own key; cab displays hold a run-scoped capability and receive their advisory as a server-sent event stream the moment it changes.
- STATION events (e.g. NTES/COA arrival/departure) update the position. Lateness of 5 minutes or more is recorded as a disruption for the controller to decide on. The feed approves nothing.
- With every involved train on a fresh live fix, recommendations become REVIEWABLE instead of PLANNING_ONLY. This is tested end to end with a signed feed simulator.
- Tested with real data: a real morning's 7,979 arrivals sent as signed STATION events at their real times; 96.6% accepted (the rest out of order or unknown runs, refused on purpose), about 0.1 s per one-minute batch. Reports are matched from each train's last observation, never from its projection.
- Remaining for go-live: CRIS's interface specification (field names, identifiers, transport), credentials, and a shadow-mode trial.

11. Legal, safety and data compliance

Instrument	How the design addresses it	Status
Railways Act 1989; General & Subsidiary Rules	Advisory only; named-controller approval; no interface to the safety chain	Met by design
RDSO / EN 50126, 50129, 50716; CRS	Outside the SIL chain by design; V&V; evidence prepared; safety case and assessor are IR steps	Needs authority
IT Act 2000 s.43/66/70 (CII)	Ingests only signed, pushed, authorised data; no scraping or probing	Met by design
CERT-In Directions 2022	Tamper-evident persistent logs (operator keeps ≥ 180 days in India), NTP-synced clocks, 6-hour incident runbook	Ready (operator config)
DPDP Act 2023 / Rules 2025	No passenger data; only controller IDs for accountability	Met by design
GODL / NDSAP, CC0, ODbL	Publisher, licence and SHA-256 recorded for every source	Met
CERT-In/STQC audit, GIGW	Self-tested; needs an empanelled auditor	Open (external)

Full register: seva2026/COMPLIANCE_REGISTER.md. This is an engineering register, not legal advice.

12. What is left

- Indian Railways / CRIS: authorise live feeds (NTES/RTIS/COA) and share the interface specification.
- Indian Railways: share block-section, loop and line-speed registers; measured on real days, they are what conflict warnings need to become precise.
- Download the current official timetable from a connection in India and ingest it (one command).
- Retrain the delay forecast on IR's own feed across seasons (fog, monsoon); it is learned from one month of real running today.
- Independent CERT-In/STQC security audit; hosting on railway infrastructure in India.
- Safety case and hazard log signed by IR; independent safety assessment; RDSO acceptance of the advisory role.
- Shadow-mode trial on one division: compare recommendations with controllers' actual decisions.

13. How to run

Task	Command (in india_rail_ai/)
Install	<code>pip install -r requirements.txt</code>
Get data and train	<code>python -m india_rail setup</code>
Real data	<code>python -m india_rail real fetch build osm --pbf FILE validate</code>
Serve the API and consoles	<code>python -m india_rail serve → /control, /control/national, /cab</code>
Tests and lint	<code>python -m pytest -q; ruff check .</code>
Simulation	<code>python -m india_rail.railguard.simulate --demo 70000 --national 90000</code>
ML training rounds	<code>python -m india_rail.training</code>
Official data	<code>python -m india_rail official --file timetable.csv</code>
All-train schedules	<code>python -m india_rail schedules --audit --train 12951 --export DIR</code>
Shadow trial	<code>POST /railguard/national/shadow/actual; GET /railguard/national/shadow/report</code>
Verify an audit log	<code>python -m india_rail.railguard.audit verify</code>
Rebuild this report	<code>python scripts/build_report.py</code>